



Xiaoji decoction induces apoptosis in lung cancer cells and M1 Macrophage polarization via the JAK1/STAT3/PD-L1 pathway

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ABSTRACT

Ethnopharmacological relevance: Xiaoji Decoction (XJD), a traditional Chinese herbal formula, has been used for decades to effectively treat non-small cell lung cancer (NSCLC), improving patients' quality of life and prolonging overall survival. Nevertheless, the molecular mechanisms responsible for the effectiveness of XJD on NSCLC remain largely unexplored.

Aim of the study: This study aims to investigate the therapeutic effects of XJD in NSCLC and to elucidate the molecular mechanisms involved.

Materials and methods: *In vitro* experiments were conducted by treating A549 cells with XJD at concentrations of 0, 6.25, 12.5, 25, and 50 mg/mL, and H1975 cells with XJD at concentrations of 0, 10, 20, 30, and 40 mg/mL. Ruxolitinib was used at a concentration of 10 μ M. MTT assays were used to evaluate cell viability, and EdU assays were employed to measure cell proliferation. Apoptosis and cell cycle progression were analyzed by flow cytometry *in vitro*. The JAK1, p-STAT3, STAT3, PD-L1, BAX, Bcl-2, Cleaved Caspase-9, and Cleaved Caspase-3 were examined through Western blot analysis. *In vivo* experiments involved allocating mice into six groups (n = 5), including a saline control group, XJD treatment group (13.4 g/kg/day by gavage), α PD-L1 group (200 mg/mouse), Ruxolitinib group (10 mg/kg/day by gavage), XJD + Ruxolitinib group, and XJD + α PD-L1 group. Animal weight and tumor size were recorded daily for two weeks, and on day 18, the animals were sacrificed for tumor and spleen analysis. The number of macrophages present in the abdominal cavity, fresh tumors, and spleen was also measured.

Results: XJD effectively inhibited cell growth, triggered apoptosis, and induced cell cycle arrest in NSCLC cells. XJD was shown to downregulate JAK1 expression, inhibit the phosphorylation of STAT3, and reduce PD-L1 protein levels. It also decreased Bcl-2 expression while increasing the levels of BAX, Cleaved Caspase-9, and

Abbreviations: BAX, Bcl-2-associated X protein; Bcl-2, B-cell lymphoma-2; BSA, bovine serum albumin; EdU, 5-ethynyl-2'-deoxy-uridine; GSDME, Gasdermin-E; ICIs, immune checkpoint inhibitors; JAK1, Janus kinase 1; LLC, Lewis lung carcinoma; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; NSCLC, Non-small cell lung cancer; PD-L1, Programmed cell death ligand 1; p-STAT3, phosphorylated signal transducer and activator of transcription 3; PVDF, polyvinylidene fluoride; RhoU, Ras Homolog Family Member U; SP1, Specificity protein 1; STAT3, signal transducer and activator of transcription 3; TCM, traditional Chinese medicine; XJD, Xiaoji Decoction; irAEs, Immune-related adverse events; WAC, WW domain-containing adaptor protein with a coiled-coil; Plk1, Polo-like kinase 1; CDKs, cyclin-dependent kinases; Cdk1, cyclin-dependent kinases 1.

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Cleaved Caspase-3, promoting cell apoptosis. Furthermore, inhibiting STAT3 phosphorylation alongside XJD treatment further amplified cell growth inhibition, induced stronger apoptosis, and more effectively arrested the cell cycle. Additionally, XJD enhanced the anti-cancer efficacy of α PD-L1, with tumor growth inhibition linked to reduced M2 macrophage infiltration and increased M1 macrophage presence. Mechanistically, combination therapies incorporating XJD with either Ruxolitinib or α PD-L1 exhibited dual therapeutic effects: 1) promotion of tumor apoptosis through enhanced caspase activation, and 2) effective suppression of the oncogenic JAK1/STAT3 signalling axis and its downstream PD-L1 expression.

Conclusions: Our results indicate that XJD suppresses NSCLC cell growth, induces cell cycle arrest, and promotes apoptosis by inhibiting the JAK1/STAT3/PD-L1 signaling pathway. XJD has been shown to enhance the anti-cancer efficacy of α PD-L1 and exerts its anti-tumor effects by modulating macrophage polarization.

1. Introduction

Lung cancer remains a significant challenge in the field of oncology, accounting for approximately 12.4 % of all new diagnosed cancers and contributing to almost 20 % of global cancer-related deaths (F. Bray et al., 2024). It remains one of the most prevalent and deadly cancers worldwide (F. Bray et al., 2018; H. Sung et al., 2021). Non-small cell lung cancer (NSCLC) is responsible for nearly four out of five lung cancer cases (A. A. Thai et al., 2021).

Diagnosing NSCLC at its early stages is challenging, as the condition frequently manifests with non-specific symptoms, resulting in most patients being diagnosed at intermediate or advanced stages (F. Kocher et al., 2015). NSCLC patients have several treatment options available, including surgical resection, radiation therapy, chemotherapy, targeted therapy, and immunotherapy (N. Duma et al., 2019). The introduction of immunotherapy, particularly immune checkpoint inhibitors (ICIs), has provided improved clinical benefits for lung cancer patients (S. J. Antonia et al., 2018; M. R. Shafique et al., 2021). However, approximately 60 %–80 % of NSCLC patients do not benefit from ICIs therapy (S. Gettinger et al., 2016; S. L. Topalian et al., 2012), primarily due to either primary or secondary drug resistance, which poses a major obstacle to cancer treatment efficacy. Therefore, there is an urgent need to explore treatments that can enhance the efficacy of ICIs.

The aberrant activation and mutation of oncogenic pathways represent a crucial aspect of tumorigenesis, progression, metastasis, and drug resistance in cancer (A. M. D. Churchhouse et al., 2024; H. J. Lee et al., 2014; M. Shen et al., 2019). It was reported that STAT3 is regulated by JAK1, which activates STAT3 after receiving signals transmitted from IL-6 and INF receptors (Y. J. Li et al., 2023). The expression of JAK1 and its activation play an important role in cancer. It is important that the JAK1 protein is phosphorylated by cytokines, growth factors and intracellular proteins to phosphorylate STAT3 (W. Jin, 2020; Y. P. Zhu et al., 2015). But some studies also have shown that compared with normal tissues, there is a marked increase in the expression and activation levels of JAK1 in lung cancer tissues, particularly in early-stage patients, which was associated with an undesirable prognosis (D. Liu et al., 2011; K. Shen et al., 2021). Meanwhile, knock-downed the expression of JAK1 significantly inhibits both the proliferation of lung cancer cells and the phosphorylation of STAT3 (S. M. Kim et al., 2012; J. Wang et al., 2021). Thus, the inhibition of JAK1 expression achieved suppression of STAT3 activation, which in turn led to the suppression of cancer cell proliferation. In lung cancer patients, STAT3 is abnormally elevated in tumor tissues, accompanied by increased PD-L1 expression (C. Liu et al., 2022). In addition, elevated STAT3 expression leads to the secretion of immunosuppressive cytokines that recruit immunosuppressive cells and induce the formation of metastatic ecotopes in lung cancer (B. Jing et al., 2020). Loss of JAK1 function and aberrant PD-L1 expression also contribute to the development of resistance to ICIs (H. Liao et al., 2023; A. Marabelle et al., 2017). Therefore, aberrant activation of the JAK1/STAT3/PD-L1 pathway is an important factor leading to resistance to ICIs, and targeting this pathway may be crucial for the treatment of lung cancer.

Extensive researches, including clinical trials and basic science studies, have demonstrated that traditional Chinese medicine (TCM) can

be beneficial as an adjuvant therapeutic option for advanced cancers, offering low toxicity. Previous research has shown that TCM formulas can improve the effectiveness of chemotherapy and different immunotherapies by regulating the immune microenvironment (C. Yang et al., 2023). In recent years, macrophages within the immune microenvironment have gained significant attention (H. Li et al., 2018b; W. P. Ma et al., 2020; Y. Yang et al., 2020; K. Zhang et al., 2023).

Xiaoji Decoction (XJD), an empirical formula developed by renowned Chinese medicine practitioner Weisheng Liu, has been used in clinical practice for the treatment of lung cancer for over two decades at the Guangdong Provincial Hospital of Chinese Medicine (Q. Yanchun & Z. Haibo, 2021). Several clinical trials and laboratory experiments have been conducted on it. The inhibitory effect of XJD on the proliferation of NSCLC cells has been demonstrated in previous experimental studies via multiple pathways, including the Akt pathway (X. S. Chai et al., 2014), the AMPK α /SP1/DNMT1 pathway (S. Zhao et al., 2016), the ERK1/2/HOTAIR/SP1/EP4 pathway (J. Wu et al., 2019), and the lncRNA-PVT1/miR181a-5P/SP1 pathway (J. Wu et al., 2020), all of which have shown efficacy both *in vitro* and *in vivo*.

Thus, Specificity protein 1 (SP1) is considered among the most significant targets for which XJD can be utilized for treating lung cancer. Studies have shown a significant correlation between elevated SP1 and STAT3 expression and their functional synergy with poor prognosis in cancer patients (C. Huang & K. Xie, 2012; E. Monteleone et al., 2019). SP1 has been demonstrated to promote Gasdermin-E (GSDME), Ras Homolog Family Member U (RhoU), genes transcription directly and to synergize with STAT3 to promote cancer cell proliferation, migration, drug resistance and angiogenesis. (C. Huang & K. Xie, 2012; J. Pan et al., 2024; Q. Xiao et al., 2017). It is evident that XJD exerts a regulatory influence over the SP1 pathway. However, the specific impact of XJD on STAT3 pathway remains to be elucidated. XJD, composed of eight herbs, its anti-tumor efficacy is not yet fully elucidated, which may be associated with multiple targets and underlying molecular mechanisms. Previous experimental evidence has shown that XJD induces apoptotic cell death in NSCLC cell lines (X. S. Chai et al., 2014; J. Wu et al., 2020); nonetheless, the precise manner in which apoptosis contributes to its anti-cancer effects remains unclear. TCM formulas have been reported to be conducive to anti-tumor treatment by regulating the immune microenvironment. It remains uncertain whether XJD's antitumor effects are mediated through immune microenvironment modulation.

In this study, we investigated the mechanisms behind XJD's inhibitory effects on lung cancer cells. Our findings revealed that XJD suppressed lung cancer cell proliferation by inducing apoptosis through the regulation of the JAK1/STAT3/PD-L1 signaling pathway, confirmed by both *in vitro* cell experiments and *in vivo* animal models. Furthermore, we examined the potential of XJD to boost the anti-cancer effects of α PD-L1. The tumor growth inhibition by XJD is linked to a decrease in M2 macrophages and an increase in M1 macrophages.

2. Materials and methods

2.1. Cell lines and cell culture

The human lung cancer cell lines A549 and H1975, and Lewis lung

carcinoma (LLC) cells were obtained from The Second Clinical Medical College of Guangzhou University of Chinese Medicine, Department of Pharmacology of Traditional Chinese Medicine. A549 cells and H1975 cells were maintained in RPMI-1640 medium, while LLC cells were cultured in DMEM. All cell cultures were supplemented with 10 % fetal bovine serum and incubated at 37 °C in a humidified incubator containing 5 % CO₂.

2.2. Preparation and quality control of Xiaoji decoction

Xiaoji Decoction (XJD) is a formula that has been approved for clinical use for over ten years at the Guangdong Provincial Hospital of Chinese Medicine, China (approval No. GDZ20080035). It is manufactured by Guangdong Kangmei Pharmaceutical Company Ltd. (Guangdong, China) and is derived from a blend of eight herbs: *Psoralea corylifolia* Linn., 15 g; *Trametes versicolor* (L.) Lloyd, 15 g; *Astragalus membranaceus* (Fisch) Bunge, 30 g; *Curcuma longa* L., 20 g; *Hedyotis diffusa* Willd., 30 g; *Buthus martensii* Karsch, 10 g; *Scolopendra subspinipes*, 15 g; and *Rheum palmatum* L., 10 g. In clinical practice, the therapeutic dosage of this formulation was standardized using the mass-to-volume ratio (w/v, g crude herbs/mL decoction) rather than solute concentration. The XJD decoction was prepared by boiling 67 g of authenticated crude herbs in 250 mL ultrapure water, yielding a w/v ratio of 0.268 g/mL. The decoction underwent sterile filtration (0.22 µm membrane) followed by vacuum evaporation to obtain a stock solution (2.5 g herb equivalents/mL). Working concentrations were subsequently prepared through dilution with specific vehicles: RPMI-1640 medium for in vitro applications (final 1 g/mL) and sterile saline solution for in vivo administration (final 1.34 g/mL). In vivo administration involved daily gavage (0.2 mL/mouse). The quality of different batches is assessed using high-performance liquid chromatography to ensure consistency. The formula's main components, preparation process, and quality control methods have been described in previous reports (X. S. Chai et al., 2014; J. Wu et al., 2020; J. Wu et al., 2019; Q. Yanchun & Z. Haibo, 2021; S. Zhao et al., 2016).

2.3. Cell viability

Human lung cancer lines A549 and H1975 cells were seeded into 96-well plates at a density of 5×10^3 cells per well. Once the cells adhered, they were treated with a series of concentration gradients of XJD for 24 and 48 h, or with different concentrations of Ruxolitinib for 48 h. Then, 20 µL of MTT solution (5 mg/mL) was added to each well, and the plates were incubated for an additional 4 h. Optical density (OD) was then measured at 490 nm using a microplate reader (Bio-Rad, USA). The experiment was performed in triplicate.

2.4. 5-Ethynyl-2'-deoxy-uridine (EdU) incorporation assay

Cell proliferation of A549 and H1975 cells was assessed using the Cell-Light™ EdU DNA cell proliferation kit. In short, A549 and H1975 cells (5×10^3) were inoculated into a 96-well plate and then incubated with the diverse concentration XJD alone or combined with Ruxolitinib treatment. After 24 h treatment, the cells were cultured into EdU with 50 µM for 2 h at 37 °C. The supernatant was removed, and the cells were fixed with 4 % paraformaldehyde in PBS for 30 min, followed by permeabilization with 0.5 % Triton X-100 for 10 min. The cells were then stained with 1 × Apollo reaction reagent for 30 min. After another permeabilization step with 0.5 % Triton X-100 for 10 min, the cells' DNA was stained with Hoechst 33342 for 30 min. Cells were then visualized under a microscope (Nikon TI2-E, Japan), with EdU-positive cells counted in at least three randomly selected fields.

2.5. Cell apoptosis analysis

Cell apoptosis was assessed using the Annexin V-FITC/PI apoptosis

kit following the manufacturer's instructions. Briefly, A549 and H1975 cells were seeded into 6-well plates at a density of 1×10^5 cells per well. Cells were then exposed to either XJD alone or in combination with Ruxolitinib for 48 h. Following treatment, cells were collected, washed twice, resuspended in 500 µL of $1 \times$ binding buffer, and incubated at room temperature in the dark with 5 µL of Annexin V-FITC and 10 µL of PI for 5 min. Apoptosis was assessed using a NovoCyte flow cytometer (ACEA Biosciences), and the percentage of apoptotic cells was quantified using Novo Express software (ACEA Biosciences).

2.6. Cell cycle analysis

The cell cycle was analyzed using the Cell Cycle Staining Kit following the manufacturer's instructions. Human lung cancer cell lines A549 and H1975 were seeded into 6-well plates at a density of 1×10^5 cells per well. The cells were then treated with either XJD alone or in combination with Ruxolitinib for 24 h. After treatment, the cells were resuspended in 1 mL of DNA staining solution and 1 µL of permeabilization solution, incubating in the dark for 5 min at room temperature. The cell cycle distribution was analyzed using a NovoCyte flow cytometer (ACEA Biosciences), and the cell cycle percentages were determined using Novo Express software (ACEA Biosciences).

2.7. Western blot analysis

The tissues from the animals and cells were lysed after treatment using RIPA buffer supplemented with a cocktail of 1 % phosphatase and protease inhibitors for 30 min on ice. The resulting suspension was centrifuged at 12,000 rpm for 15 min at 4 °C, and the supernatant was collected. Protein concentration in the total cell lysate was assessed using the BCA protein assay kit with bovine serum albumin (BSA) as a standard. The samples were heated at 100 °C for 10 min prior to analysis. Equal amounts of protein (40 µg) were loaded onto a Precast Protein Plus Gel using a vertical electrophoresis system (Bio-Rad, USA) and transferred onto a polyvinylidene fluoride (PVDF) membrane. The membrane was then blocked with 5 % BSA in TBST solution for 1 h. After blocking, the PVDF membrane was washed with TBST. Subsequently, it was incubated overnight at 4 °C with primary antibodies (Supplementary Table 3.). After washing, the membrane was incubated with a secondary antibody at room temperature for 1 h. The PVDF membranes were washed and then exposed to freshly prepared ECL solution. The signals were detected using the ChemiDoc XRS + Imaging System (Bio-Rad, USA). Finally, densitometry analysis was performed using ImageJ software (National Institutes of Health, USA).

2.8. In vivo experiments

Male C57BL/6J mice (6–8 weeks old) were obtained from Guangzhou Shijia Jingda Biotechnology Co., Ltd., and were housed in the Experimental Animal Center of Guangdong Provincial Hospital of Chinese Medicine. All in vivo experiments were approved by the Institutional Animal Care and Use Committee of Guangdong Provincial Hospital of Chinese Medicine (No. 2022105), and rigorous animal studies were conducted to ensure the highest animal welfare standards and ethical conduct in research. The mice were randomly divided into six groups. In the subcutaneous model, each mouse received an injection of 2×10^6 LLC cells suspended in cold PBS. The injection was administered on the right side of its back, with a total volume of 0.2 mL. After 7 days, when the subcutaneous tumors became palpable, the mice were allocated into six groups (n = 5), including saline (control), XJD (13.4 g/kg/day by gavage), αPD-L1 (200 mg/mouse), Ruxolitinib (10 mg/kg/day by gavage), XJD + Ruxolitinib, and XJD + αPD-L1. Animal weight and tumor size were measured daily for two weeks using a digital caliper. Tumor size was calculated using the formula $(D^2 \times d)/2$, where D is the larger diameter and d is the smaller diameter of the tumor. On day 18, the mice were euthanized, and the tumors were removed.

Primary cells were extracted from the abdominal cavity, freshly excised tumors, and spleen for immunological cell analysis via flow cytometry, following previously established methods.

In clinical treatment, adults are required to ingest 100 mL of the prescription daily, with each 100 mL containing 67 g of crude drug. The formula employed to convert a human dose to a mouse dose is calculated based on body surface area: Mouse dose (g/kg) = Human dose (g/kg) × 37/3. This calculation indicates that the mouse requirement to ingest Xiaoji Decoction is approximately 13.4 g/kg. Moreover, as indicated by earlier research, Xiaoji Decoction has been shown to exert a prominent inhibitory effect on tumor growth in mouse models, particularly when administered at medium (13.4 g/kg) and high (26.8 g/kg) doses (S. Zhao et al., 2016). Also, the medium dose of Xiaoji Decoction (13.4 mg/kg), when combined with Gefitinib (J. Wu et al., 2019) or Cisplatin (J. Wu et al., 2020) has been demonstrated to markedly suppress tumour growth in mouse models. Based on the 3Rs (Refinement, Reduction, and Replacement) principles, the animal experiment was conducted single dose (13.4 mg/kg).

2.9. Flow cytometry

To isolate immunological cells from the abdominal cavity, 5 mL of PBS was injected into the peritoneum of euthanized mice, which were then turned upside down several times to ensure thorough mixing. The PBS was then collected from the abdominal cavity. For tissue samples, the spleen or tumors were finely chopped and then digested with a Collagenase I solution (0.8 mg/mL) in a shaking incubator at 37 °C for 30 min. After digestion, the tissue samples were filtered through a 70 µm cell filter to eliminate larger debris. The resulting suspension was centrifuged to collect primary cells. Spleen primary cells were treated with 1 × ACK buffer for 5 min to lyse red blood cells, after which lysis was stopped, and the cells were washed with PBS. The cell suspension was incubated with antibodies in the dark for 30 min, then washed again with PBS and analyzed using the NovoCyte Quanteon (Agilent, USA). Flow cytometry data were processed using Novo Express software (ACEA Bio-sciences).

2.10. Statistical analysis

Data were presented as mean ± Standard Deviation (SD). Statistical analysis was performed using SPSS 27.0 software. One-way analysis of variance and the Kruskal-Wallis test were applied for pairwise comparisons, with $p < 0.05$ considered statistically significant.

3. Results

3.1. XJD inhibited growth of NSCLC in vitro

To examine the impact of XJD on cell viability and proliferation in NSCLC, A549 and H1975 cell lines were exposed to different concentrations of XJD for 24 or 48 h. The MTT assay demonstrated that XJD reduced the viability of lung cancer cells in a time- and dose-dependent manner (Fig. 1A–B). Notably, XJD concentrations of 50 mg/mL in A549 cells and 40 mg/mL in H1975 cells significantly inhibited cell viability after 48 h, prompting us to select concentrations of 0, 6.25, 12.5, 25, and 50 mg/mL for A549 cells and 0, 10, 20, 30, and 40 mg/mL for H1975 cells in subsequent experiments. To further validate the effects of XJD on cell proliferation, an EdU assay was performed. The results of the EdU assay (Fig. 1C–F) indicated that XJD decreased the number of EdU-positive cells in both A549 (Fig. 1C–D) and H1975 (Fig. 1E–F) lines. To investigate the mechanism by which XJD inhibits NSCLC cell growth, we analyzed cell cycle distributions in A549 and H1975 cells treated with varying doses of XJD using flow cytometry. Our findings revealed that XJD inhibited the transition from the G2 to G1 phase in A549 cells (Fig. 1G–H) and induced cell cycle arrest in the G1 phase in H1975 cells, thereby preventing progression to the S phase (Fig. 1I–J). In summary,

these results indicate that XJD effectively suppresses the proliferation of NSCLC cells.

3.2. XJD promoted apoptosis of NSCLC

To guide further experimental investigation into XJD's mechanism, we first employed network pharmacology and molecular docking. This approach identified 22 potential anti-lung cancer components, with five core compounds – quercetin, kaempferol, formononetin, aloe-emodin, and 7-O-methylisomucronulatol – prioritized for subsequent validation due to their roles in apoptosis induction and immunomodulation. Target analysis revealed 107 shared XJD – lung cancer targets, of which IL6, AKT1, BCL2, BAX, and HSP90AA1 were central hubs (Fig. S1). Critically, enrichment analysis confirmed these targets were significantly associated with apoptosis regulation, IL-17, and TNF pathways, aligning with the JAK1/STAT3/PD-L1 axis.

Apoptosis is recognized as a key anti-proliferative mechanism of anticancer drugs. To further explore the anti-NSCLC effects of XJD, we examined its role in inducing apoptosis in the cells. The apoptosis assay showed a significant rise in the rate of apoptotic cells in both A549 (Fig. 2A–B) and H1975 (Fig. 2C–D) cell lines after treatment with XJD. Subsequently, the levels of apoptotic proteins were analyzed using Western blot assay, revealing that XJD treatment triggered a dose-dependent apoptotic response in A549 (Fig. 2E–I) and H1975 (Fig. 2J–N) cells. This response was characterized by increased expression of Bax, Cleaved Caspase-9, and Cleaved Caspase-3, alongside a reduction in Bcl-2 expression in NSCLC cells (Fig. 2E–N). Overall, these findings suggest that XJD effectively induces apoptosis in NSCLC cells.

3.3. XJD significantly inhibited the expression of JAK1, p-STAT3, and PD-L1 in NSCLC cells

The potential molecular mechanisms underlying XJD's inhibitory effects on cell growth were also explored. Considering the crucial roles of JAK1, p-STAT3, and PD-L1 in tumor proliferation, we examined the impact of XJD on the expression of these proteins in A549 (Fig. 3A–D) and H1975 (Fig. 3E–H) cells using Western blotting. The results revealed that XJD significantly suppressed the expression of JAK1, p-STAT3, and PD-L1 in a dose-dependent manner (Fig. 3A–H).

3.4. STAT3 phosphorylation is involved in the inhibition of A549 and H1975 cell proliferation by XJD

To investigate whether the reduction of p-STAT3 expression affects the proliferation and apoptosis of NSCLC, a JAK/STAT inhibitor Ruxolitinib (INCB018424) was selected to treat the cells. A549 (Fig. 4A) and H1975 (Fig. 4B) cells were treated with various concentrations of Ruxolitinib for 48 h. The MTT assay revealed that Ruxolitinib reduced the viability of both cell lines in a dose-dependent manner (Fig. 4A–B). These results suggested a strong link between STAT3 phosphorylation and the proliferation of NSCLC cells, highlighting that targeting STAT3 could be an effective strategy for lung cancer treatment. Notably, Ruxolitinib at 10 µM significantly reduced cell viability in A549 cells ($p < 0.05$), whereas a tendency to decrease no significant inhibition was observed in H1975 cells (Fig. 4B). Based on the pronounced effect in A549 cells and prior evidence of Ruxolitinib's dose-dependent activity in NSCLC models, we selected 10 µM for subsequent experiments, alongside XJD at 30 mg/mL for A549 cells and 20 mg/mL for H1975 cells.

To examine the impact of p-STAT3 suppression on the inhibitory effects of XJD, MTT and EdU assays were conducted. The results showed that inhibiting p-STAT3 significantly reduced both the proliferation rate and the number of EdU-positive cells in A549 (Fig. 4E–G) and H1975 (Fig. 4F–H) cells treated with XJD. The MTT assays indicated that XJD exerted an even stronger inhibitory effect on NSCLC cells following the suppression of p-STAT3 expression. The combination of XJD and

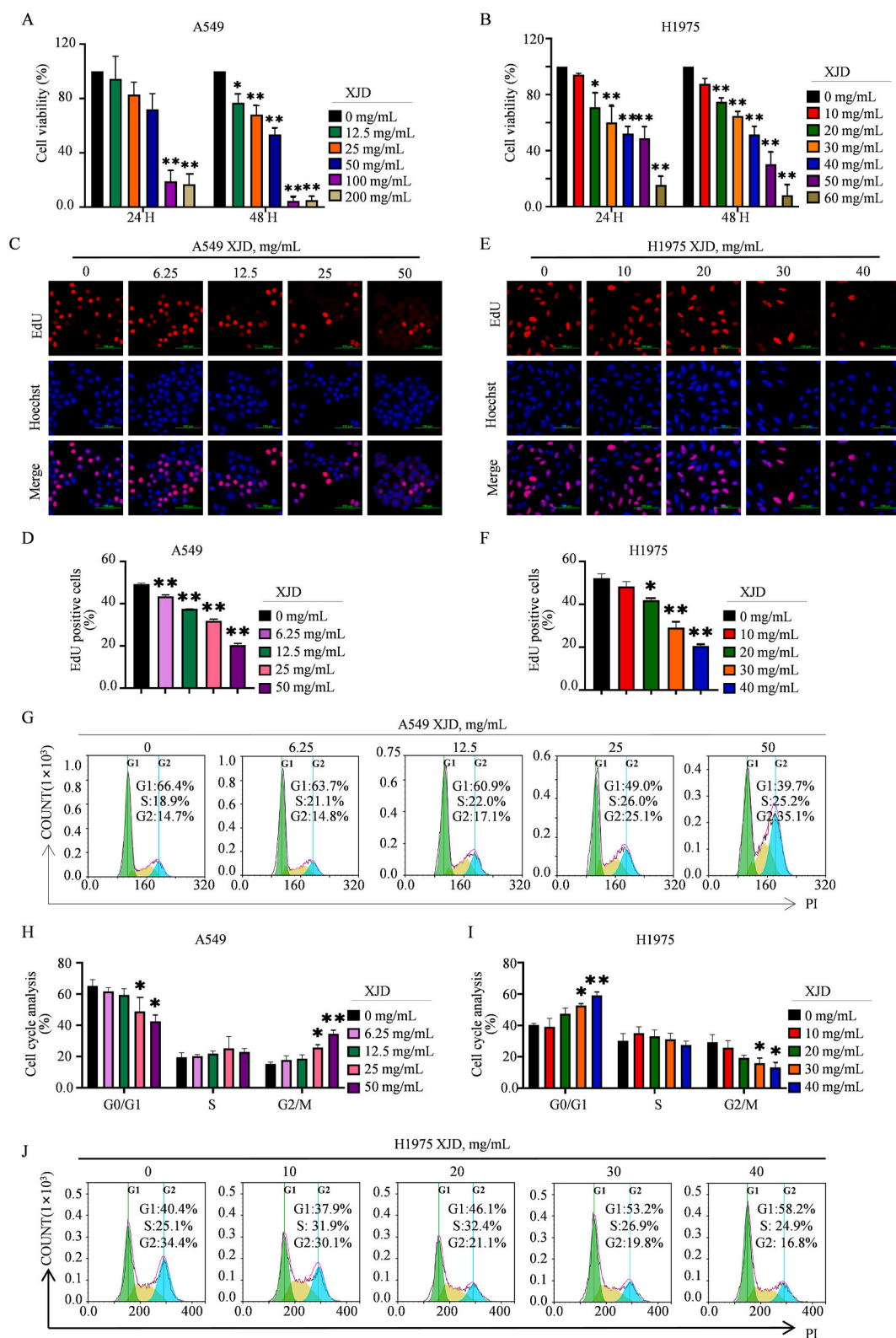
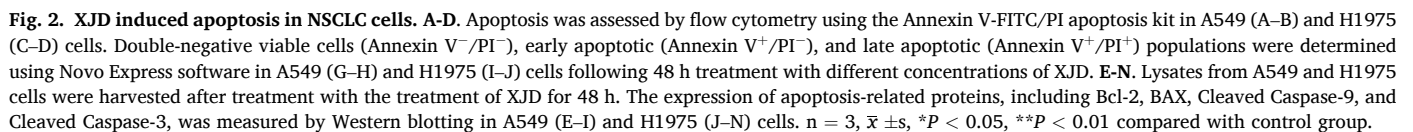


Fig. 1. XJD suppressed cell growth and induces cell cycle arrest in NSCLC cells. **A–B.** The impact of XJD on cell viability was assessed in A549 (**A**) and H1975 (**B**) cells treated with varying concentrations of XJD for up to 24 h and 48 h, as measured by MTT assays. **C–F.** Cell proliferation in A549 (**C–D**) and H1975 (**E–F**) cells was evaluated using the EdU assay following 24 h treatment with XJD. After staining, cells were visualized under an inverted microscope, and EdU-positive cells were manually counted in at least three randomly selected fields per well. **G–J.** Flow cytometry was utilized to analyze cell cycle distribution, and the percentages of each phase (G0/G1, S, and G2/M) were determined using Novo Express software in A549 (**G–H**) and H1975 (**I–J**) cells following 24 h treatment with XJD. Data are presented as mean $\pm$ standard deviation ($n = 3$). * $P < 0.05$, ** $P < 0.01$ compared with control group.



Additionally, cell cycle analysis revealed that XJD effectively reduced the number of cells in the G0/G1 phase while increasing the number in the G2/M phase in A549 cells. Conversely, it increased the

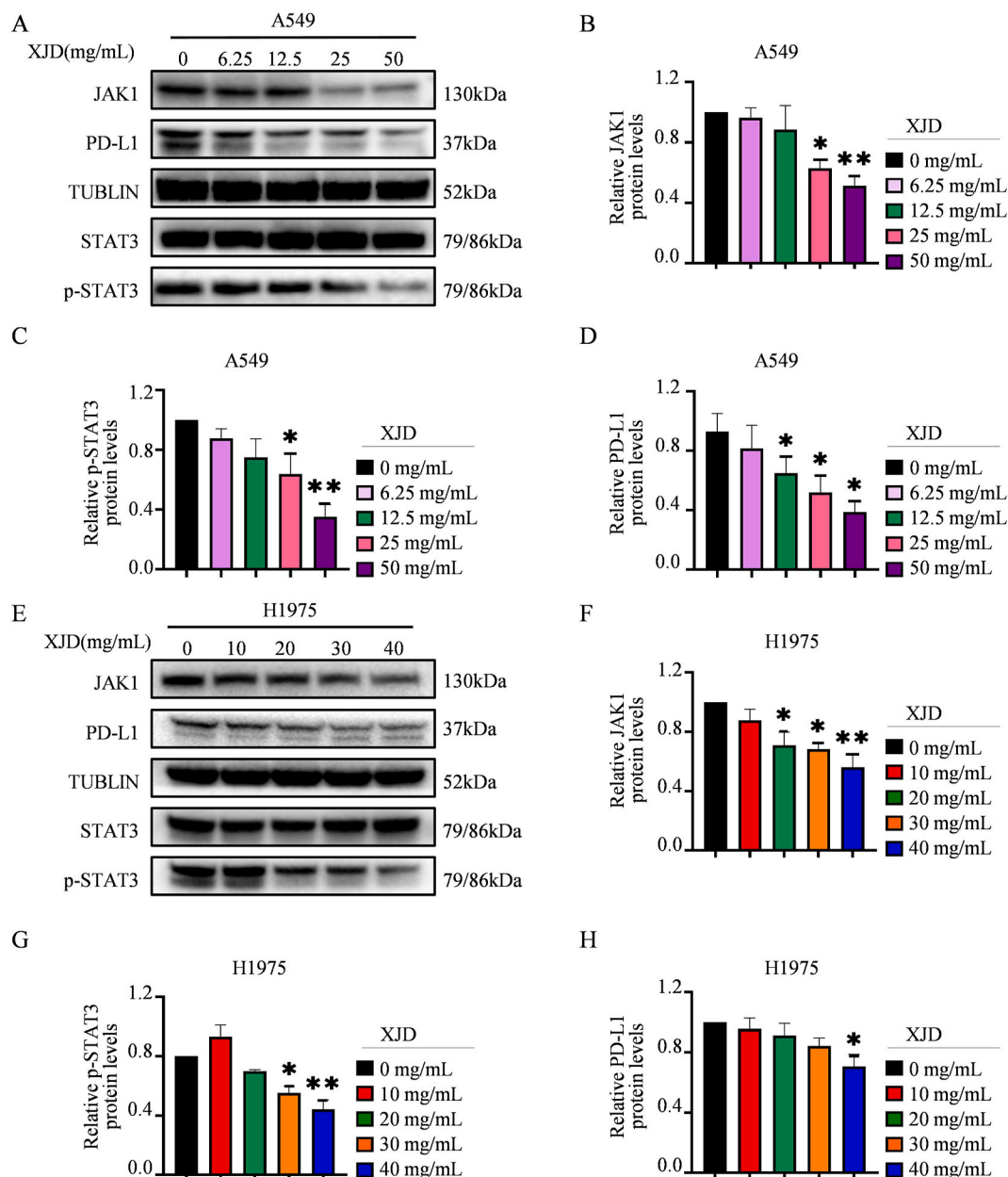


Fig. 3. XJD inhibited the expression of JAK1/STAT3/PD-L1 signaling in NSCLC cells. A-H. Lysates from A549 and H1975 cells were harvested after 48 h of treatment with different concentrations of XJD. Western blot analysis was performed to measure the expression of JAK1, STAT3, p-STAT3, and PD-L1 in A549 (A-D) and H1975 (E-H) cells. *n* = 3, $\bar{x} \pm s$, **P* < 0.05, ***P* < 0.01 compared with control group.

G0/G1 phase cells and decreased the G2/M phase cells in H1975 cells when p-STAT3 was inhibited. Interestingly, the combination treatment of XJD and Ruxolitinib increased the number of G2/M phase cells in A549 cells (Fig. 4I-J) and G0/G1 phase cells in H1975 cells (Fig. 4K-L) compared to the XJD group. Overall, these findings suggest that by suppressing p-STAT3, XJD enhances its inhibitory effect on the proliferation of NSCLC cells.

3.5. STAT3 phosphorylation is involved in the induction of A549 and H1975 cell apoptosis by XJD

Results from earlier experiments indicated that XJD influences the expression of apoptotic proteins and promotes apoptosis. To evaluate whether the inhibition of JAK/STAT3 pathway affects the apoptotic effects induced by XJD, apoptosis assay and Western blot analysis were conducted. Ruxolitinib was selected to combine with XJD. The

combination of XJD and Ruxolitinib showed a significant increase in apoptosis induction in both A549 and H1975 cells compared to the XJD group (Fig. 5A-D).

Additionally, the effects of XJD combined with Ruxolitinib on the expression of Bcl-2, BAX, Cleaved Caspase-9, and Cleaved Caspase-3 in A549 and H1975 cells were further analyzed. The combination of XJD with Ruxolitinib led to a significant rise in the expression of BAX, Cleaved Caspase-9, and Cleaved Caspase-3, along with a marked reduction in Bcl-2 levels in H1975 cells, compared to the XJD group alone (Fig. 5E-N). In A549 cells, Bcl-2 was decreased and Cleaved Caspase-3 was increased in combination group compared to XJD group. Overall, these results suggest that by suppressing p-STAT3, XJD enhances its inhibitory effect on the proliferation of NSCLC cells.

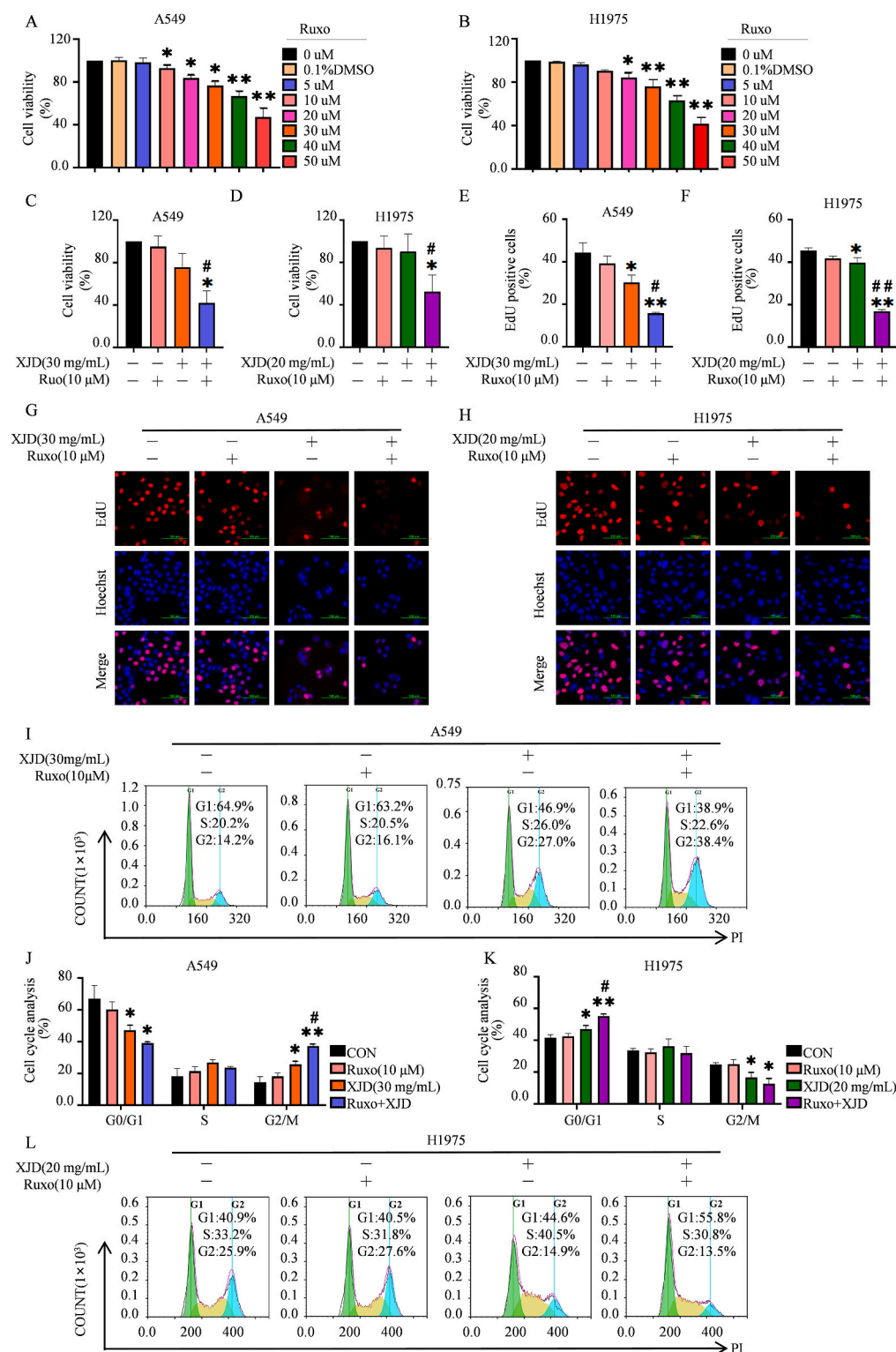


Fig. 4. The effects of XJD on the proliferation of NSCLC cells were analyzed after suppression of p-STAT3. A-B. The impact of the STAT3 inhibitor, Ruxolitinib, on cell viability was assessed in A549 (A) and H1975 (B) cells subjected to various concentrations of Ruxolitinib for up to 48 h. Cell viability was evaluated using MTT assays. C-D. A549 cells were treated with XJD (30 mg/mL) and Ruxolitinib (10 μ M), and H1975 cells were treated with XJD (20 mg/mL) and Ruxolitinib (10 μ M) for 48 h. The effects of XJD on the proliferation of A549 (C) and H1975 (D) cells were observed after suppression of p-STAT3, using MTT assays. E-H. A549 cells were treated with XJD (30 mg/mL) and Ruxolitinib (10 μ M), and H1975 cells were treated with XJD (20 mg/mL) and Ruxolitinib (10 μ M) for 24 h. The EdU assay assessed cell proliferation to explore how XJD works, after the suppression of p-STAT3 in A549 (E, G) and H1975 (F, H) cells. I-L. A549 cells were treated with XJD (30 mg/mL) and Ruxolitinib (10 μ M), and H1975 cells were treated with XJD (20 mg/mL) and Ruxolitinib (10 μ M) for 24 h. The effect of XJD on cell cycle distribution after inhibition of p-STAT3 in A549 (I-J) and H1975 (K-L) cells was measured using flow cytometry. The percentages of each phase (G0/G1, S, and G2/M) were determined using Novo Express software. $n = 3$, $\bar{x} \pm s$, * $P < 0.05$, ** $P < 0.01$ compared with control group, # $P < 0.05$, ## $P < 0.01$ compared with XJD group.

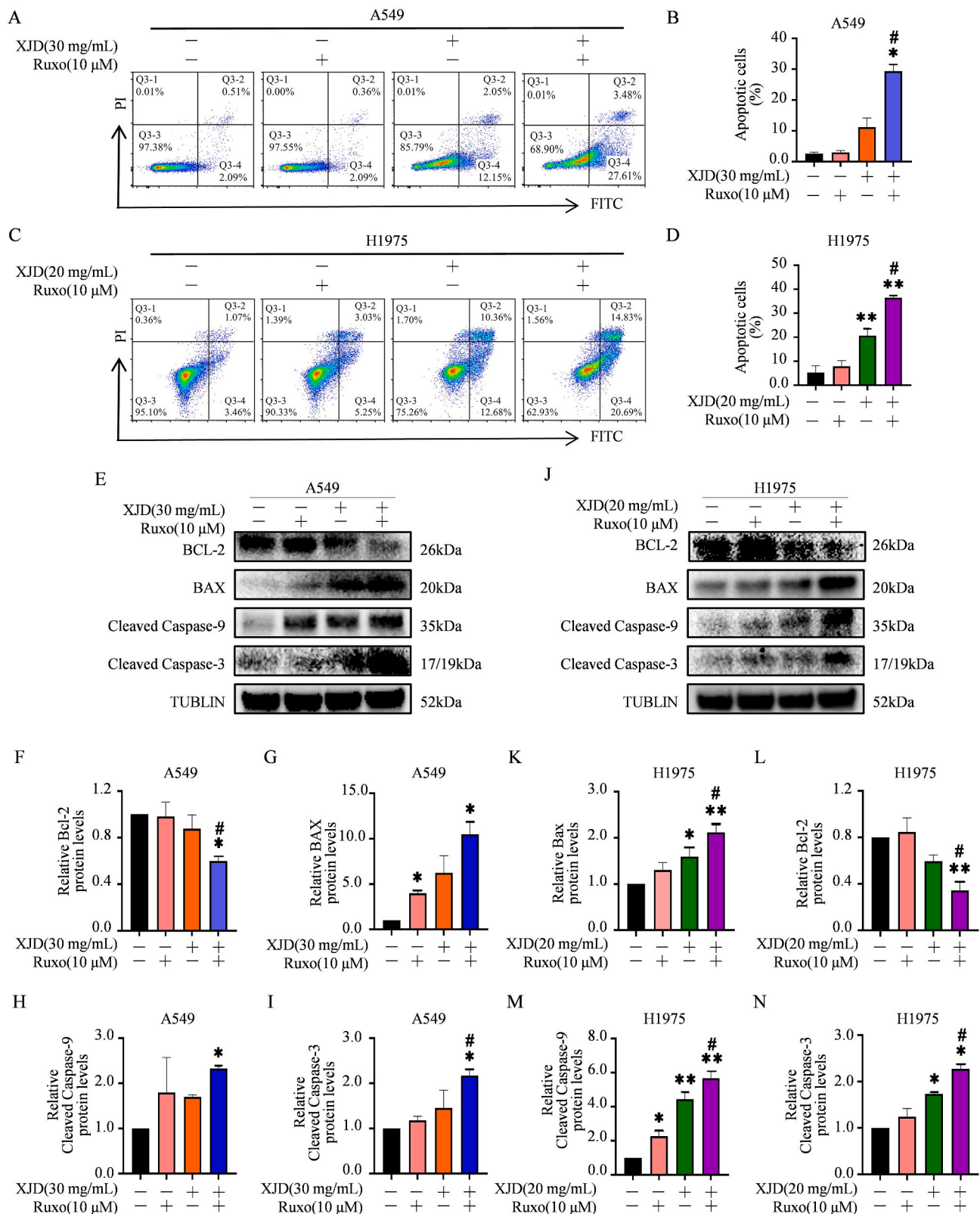


Fig. 5. The effects of XJD on apoptosis in A549 and H1975 cells were observed after suppressing p-STAT3. A549 cells were treated with XJD (30 mg/mL) and Ruxolitinib (10 μM), and H1975 cells were treated with XJD (20 mg/mL) and Ruxolitinib (10 μM) for 48 h. **A–D.** After treatment, the Annexin V-FITC/PI apoptosis kit was used to analyze apoptosis in A549 (**A–B**) and H1975 (**C–D**) cells by flow cytometry. The populations of double-negative viable cells (Annexin V⁻/PI⁻), early apoptotic cells (Annexin V⁺/PI⁻), and late apoptotic cells (Annexin V⁺/PI⁺) were quantified using Novo Express software in A549 (panels **G–H**) and H1975 (panels **I–J**) cells. **E–N.** Lysates from A549 and H1975 cells were harvested after treatment with the treatment for 48 h. The expression of apoptosis-related proteins, including Bcl-2, BAX, Cleaved Caspase-9, and Cleaved Caspase-3, was measured by Western blotting in A549 (**E–I**) and H1975 (**J–N**) cells. $n = 3$, $\bar{x} \pm s$, * $P < 0.05$, ** $P < 0.01$ compared with control group; # $P < 0.05$, ## $P < 0.01$ compared with XJD group.

3.6. The effects of XJD on JAK1 and PD-L1 expression of A549 cells and H1975 cells were observed after suppression of p-STAT3

Considering the essential role of JAK1/STAT3/PD-L1 signaling in tumor development, we further explored the effects of XJD combined with Ruxolitinib on the expression levels of JAK1, p-STAT3, and PD-L1 in A549 and H1975 cells using Western blot assay. The findings showed that this combination had no significant effect on JAK1 expression in either cell line (Fig. 6A–C, F). However, it did have a significant impact on the levels of p-STAT3 and PD-L1 proteins (Fig. 6A–B, D–E, G–H). Overall, these results highlighted the crucial role of STAT3 in mediating the inhibitory effects of XJD on cell growth and apoptosis.

3.7. XJD combined with α PD-L1 alleviates effect on tumorigenesis in C57BL/6 mice

The XJD has been demonstrated to markedly suppressed tumor growth in nude mouse xenograft model, with a clear dose-dependent effect (S. Zhao et al., 2016). Immune checkpoint inhibitors (ICIs)

targeting PD-L1, such as atezolizumab and pembrolizumab, are clinically approved for NSCLC and have shown significant survival benefits in pivotal trials (C. Weinstock et al., 2017). Recent studies have demonstrated that STAT3 critically regulates PD-L1 expression through transcriptional control, a mechanism central to adaptive immune resistance in NSCLC. Furthermore, while α PD-L1 therapies improve outcomes, they are limited by immune-related adverse events (irAEs) (V. A. Nikolaou et al., 2022; Q. Zhu et al., 2025; S. Zhu et al., 2021). Pre-clinical studies suggest that traditional Chinese medicines like XJD may mitigate irAEs while enhancing antitumor immunity through complementary pathways. By evaluating XJD in combination with α PD-L1, this study aims to explore synergistic mechanisms that could address these limitations, potentially offering safer and more effective therapeutic strategies for NSCLC. In this study, the therapeutic efficacy of XJD was evaluated using a subcutaneous mouse model of LLC cells in this study. The results clearly demonstrated that the combination of XJD and Ruxolitinib group, as well as the combination of XJD and α PD-L1 group, significantly suppressed the tumor growth (Fig. 7A–C). Concomitantly, there were no notable alterations in body weight among the

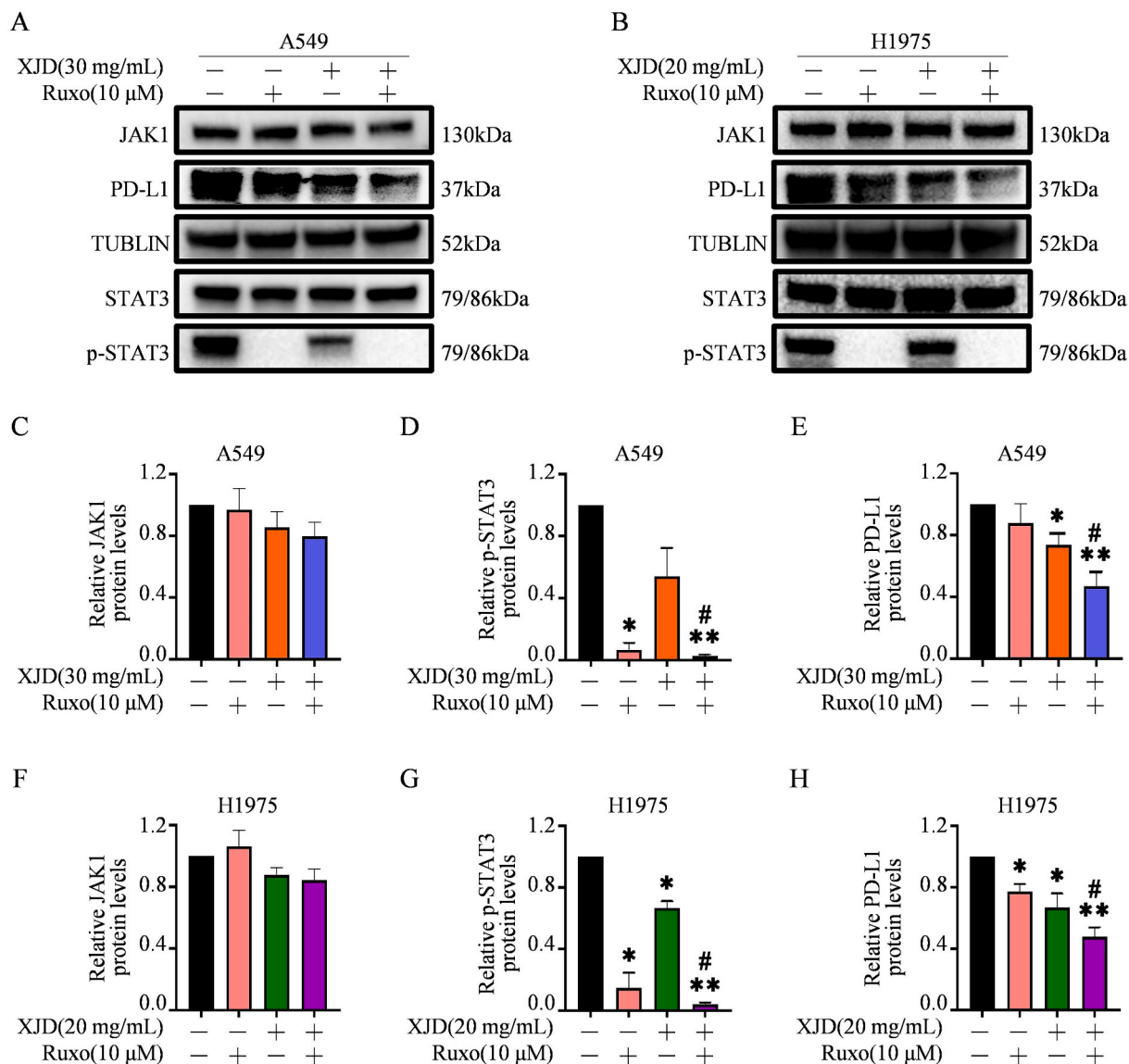
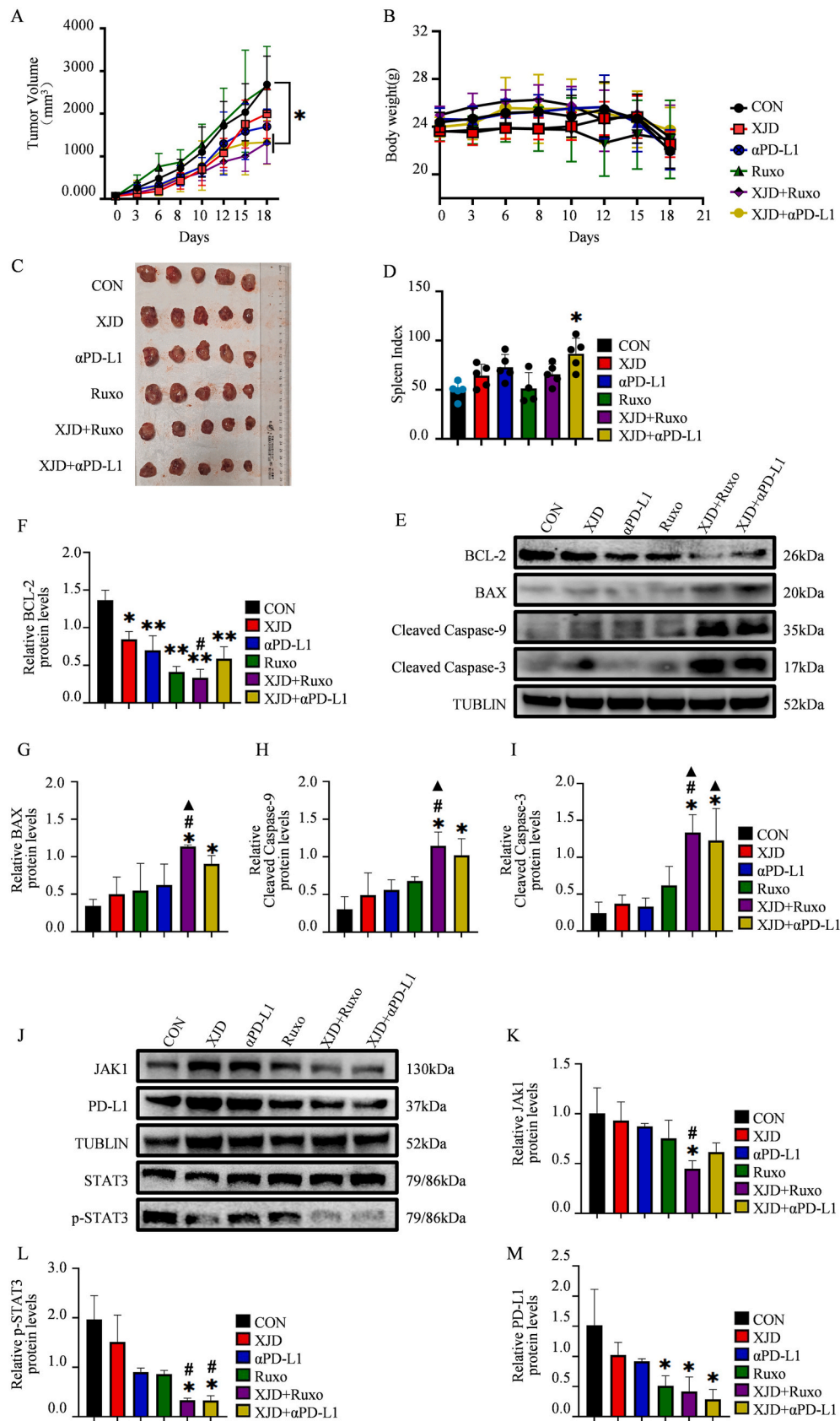


Fig. 6. The effects of XJD on JAK1 and PD-L1 expression of A549 cells and H1975 cells were observed after suppression of p-STAT3. A-H. A549 cells were treated with XJD (30 mg/mL) and Ruxolitinib (10 μ M), and H1975 cells were treated with XJD (20 mg/mL) and Ruxolitinib (10 μ M) for 48 h. Lysates from A549 and H1975 cells were harvested after 48 h of treatment. Western blot analysis was performed to measure the expression of JAK1, STAT3, p-STAT3, and PD-L1 in A549 cells (A, C-E) and H1975 cells (B, F-H). $n = 3$, $\bar{x} \pm s$; * $P < 0.05$, ** $P < 0.01$ compared with control group; # $P < 0.05$, ## $P < 0.01$ compared with XJD group.



(caption on next page)

Fig. 7. XJD combined with α PD-L1 alleviates effect on tumorigenesis in C57BL/6 mice. Each mouse received an injection of 2×10^6 LLC cells on the right side of its back. After 7 days, when the subcutaneous tumors became palpable, the mice were randomly allocated into six groups: Con (saline/day by gavage), XJD (13.4 g/kg/day by gavage), α PD-L1 (200 mg/mouse every 3 days by injection), Ruxolitinib (10 mg/kg/day by gavage), XJD (13.4 g/kg/day by gavage) + Ruxolitinib (10 mg/kg/day by gavage), and XJD (13.4 g/kg/day by gavage) + α PD-L1 (200 mg/mouse every 3 days by injection). Each group consisted of five mice. Animal weight and tumor size were measured. Tumor size was calculated using the formula $(D^2 \times d)/2$, where D is the larger diameter and d is the smaller diameter of the tumor. On day 18, the mice were euthanized, and the tumors and spleen were removed. **A-B.** The volume of tumors (A) and body weight of the LLC model (B) were measured, $n = 5$. **C.** Images of tumors removed from the LLC model, $n = 5$. **D.** Spleen index of the LLC model, $n = 4-5$. At the end of the experiments, tumors were removed from each animal. Lysates from the tumor tissues were harvested, and the expression of apoptosis-related proteins, including Bcl-2 (E, F), BAX (E, G), Cleaved Caspase-9 (E, H) and Cleaved Caspase-3 (E, I), was measured by Western blotting, $n = 3$. **J-M.** Western blotting quantified the JAK1/STAT3/PD-L1 signalling pathway in tumour tissues from the LLC model, $n = 3$. $\bar{x} \pm s$; * $P < 0.05$, ** $P < 0.01$ compared with control group; # $P < 0.05$, ## $P < 0.01$ compared with XJD group; $\blacktriangle P < 0.05$, $\blacktriangle\blacktriangle P < 0.01$ compared with α PD-L1 group.

experimental groups (Fig. 7B). To gain further insight into the effect of XJD on the immune system, the spleen index was measured. The results demonstrated that the spleen index was significantly increased in the XJD combined with α PD-L1 treatment group compared to the control group (Fig. 7D).

3.8. XJD combined with α PD-L1 induced apoptosis and suppressed JAK1/STAT3/PD-L1 signaling pathway *in vivo*

In this study, we have shown that XJD effectively inhibits tumor cell proliferation and promotes apoptosis, which correlates with the suppression of the JAK1/STAT3/PD-L1 signaling pathway *in vitro*. Changes in proteins associated with tumor apoptosis *in vivo* were assessed using Western blot assays. The levels of the Bcl-2 protein were significantly reduced in the treated groups (Fig. 7E–F), while the expression of Bax (Fig. 7E, G), Cleaved Caspase-9 (Fig. 7E–H), and Cleaved Caspase-3 (Fig. 7E–I) was elevated compared to the control group. Notably, the levels of Bax (Fig. 7E, G), Cleaved Caspase-9 (Fig. 7E–H), and Cleaved Caspase-3 (Fig. 7E–I) were significantly elevated in the groups receiving both XJD and Ruxolitinib treatment, as well as in those treated with both XJD and α PD-L1, compared to the groups treated with XJD or α PD-L1 alone (Fig. 7E–G–I). Regarding the JAK1/STAT3/PD-L1 signaling pathway, Western blot analysis showed a significant reduction in the protein levels of JAK1 (Fig. 7J, K), p-STAT3 (Fig. 7J–L), and PD-L1 (Fig. 7J–M) in the combined treatment groups when compared to the control group or the XJD group (Fig. 7J–M).

3.9. XJD promotes M1 macrophages polarized and inhibits M2 macrophages polarized *in vivo*

Macrophages in the immune microenvironment play a crucial role in tumorigenesis, tumor progression, and metastasis. Some clinical studies have shown that patients treated with XJD have changes in the number of lymphocytes. To further investigate the impact of XJD on macrophages, macrophages from abdominal cavity, spleen, and tumors were measured through flow cytometry. The results indicated that there are no significant changes in M1 macrophages in the abdominal cavity among the groups (Fig. 8A–D). The levels of splenic M1 macrophages were increased in the α PD-L1 group, the XJD and Ruxolitinib co-treatment group, and the XJD and α PD-L1 co-treatment group compared to the control group after treatment (Fig. 8B–E). In tumor tissues, M1 macrophages were significantly elevated in the treated groups compared to the control group, with the highest levels observed in the XJD and Ruxolitinib co-treatment group and the XJD and α PD-L1 co-treatment group compared to the XJD or α PD-L1 group (Fig. 8C, F). M2 macrophages in the abdominal cavity were notably reduced in the α PD-L1 group, the XJD and Ruxolitinib co-treatment group, and the XJD and α PD-L1 co-treatment group compared to the control group or XJD group (Fig. 8G–J). Splenic M2 macrophages decreased in the XJD and α PD-L1 co-treatment group compared to the control group and the XJD group after treatment (Fig. 8H–K). Only minor changes in M2 macrophages were observed in the tumor across all groups (Fig. 8I–L). To validate these *in vivo* observations, *in vitro* polarization assays using THP-1 cells demonstrated that XJD directly inhibits M2 macrophage

differentiation (Fig. S2). Collectively, these findings confirm XJD's capacity to modulate macrophage plasticity, with mechanistic differences observed between systemic immunosuppressive environments (as seen in peritoneal and splenic compartments) and direct cellular effects (evidenced by *in vitro* assays), while tumor-associated M2 macrophages resistance requires further mechanistic exploration.

4. Discussion

The principles of TCM advocate for tailored pharmacological interventions based on a comprehensive understanding of the patient's cause and mechanism of disease. In the context of cancer therapy, these approaches aim to facilitate the coexistence of both the patient and tumor cells, while also impeding tumor growth (C. Yongzhong & Z. Ruisheng, 2022). These approaches focus on extending the patient's lifespan while enhancing their overall physical condition, boosting the immune system, and minimizing the occurrence and severity of side effects associated with conventional cancer treatments such as chemotherapy and radiation therapy.

In recent years, the use of TCM as an adjuvant in cancer treatment has attracted significant attention, making the study of TCM's mechanisms of action against cancer a key area of research. The anti-cancer effects of TCM can be broadly categorized into two main areas: the direct targeting of cancer cells and the indirect influence on the surrounding microenvironment. TCMs have been observed to regulate the expression of oncogenic pathways (A. M. D. Churchhouse et al., 2024; X. Yang et al., 2021), induce the death of cancer cells (including apoptosis (X. B. Jia et al., 2021), pyroptosis (D. Luo et al., 2023), ferroptosis (Z. Zhang et al., 2022), and necroptosis (Z. Feng et al., 2024)), and inhibit cancer cell proliferation, invasion, metastasis, and the development of tumor drug resistance (M. Yu et al., 2017). Additionally, TCMs have been demonstrated to inhibit tumor angiogenesis (Y. Peng et al., 2018; G. Zou et al., 2020) and release suppression of the immune microenvironment (Z. Feng et al., 2024), thereby creating an unfavorable environment for tumor cell survival.

Previous studies have suggested that XJD may inhibit tumor growth both *in vivo* and *in vitro* (X. S. Chai et al., 2014; J. Wu et al., 2020; J. Wu et al., 2019; S. Zhao et al., 2016). It has been indicated that XJD elicits apoptosis in non-small cell lung cancer (NSCLC) cell lines (X. S. Chai et al., 2014; J. Wu et al., 2020); The TP53 gene has gained widespread recognition for its crucial role in the regulation of apoptotic and cell cycle pathways (Green & Chipuk, 2006). To investigate the role of apoptosis in XJD's anti-tumor mechanism, we selected the A549 and H1975 NSCLC cell lines, which differ in TP53 status. Prior characterization of these cell lines revealed that H1975 cells harbor a TP53 mutation, whereas A549 cells exhibit wild-type (WT) TP53 expression (M. Nenkov et al., 2020). In the current experiment, *in vitro* analyses showed that XJD inhibited the proliferation of A549 and H1975 cells, which was associated with decreased cellular viability and EdU positivity following treatment with various concentrations of XJD. Additionally, XJD was found to disrupt the cell cycle in lung cancer cells. These results align with those of earlier experimental studies (J. Wu et al., 2019). Overall, the results indicate that XJD inhibits lung cancer cell growth and demonstrates anti-tumor effects.

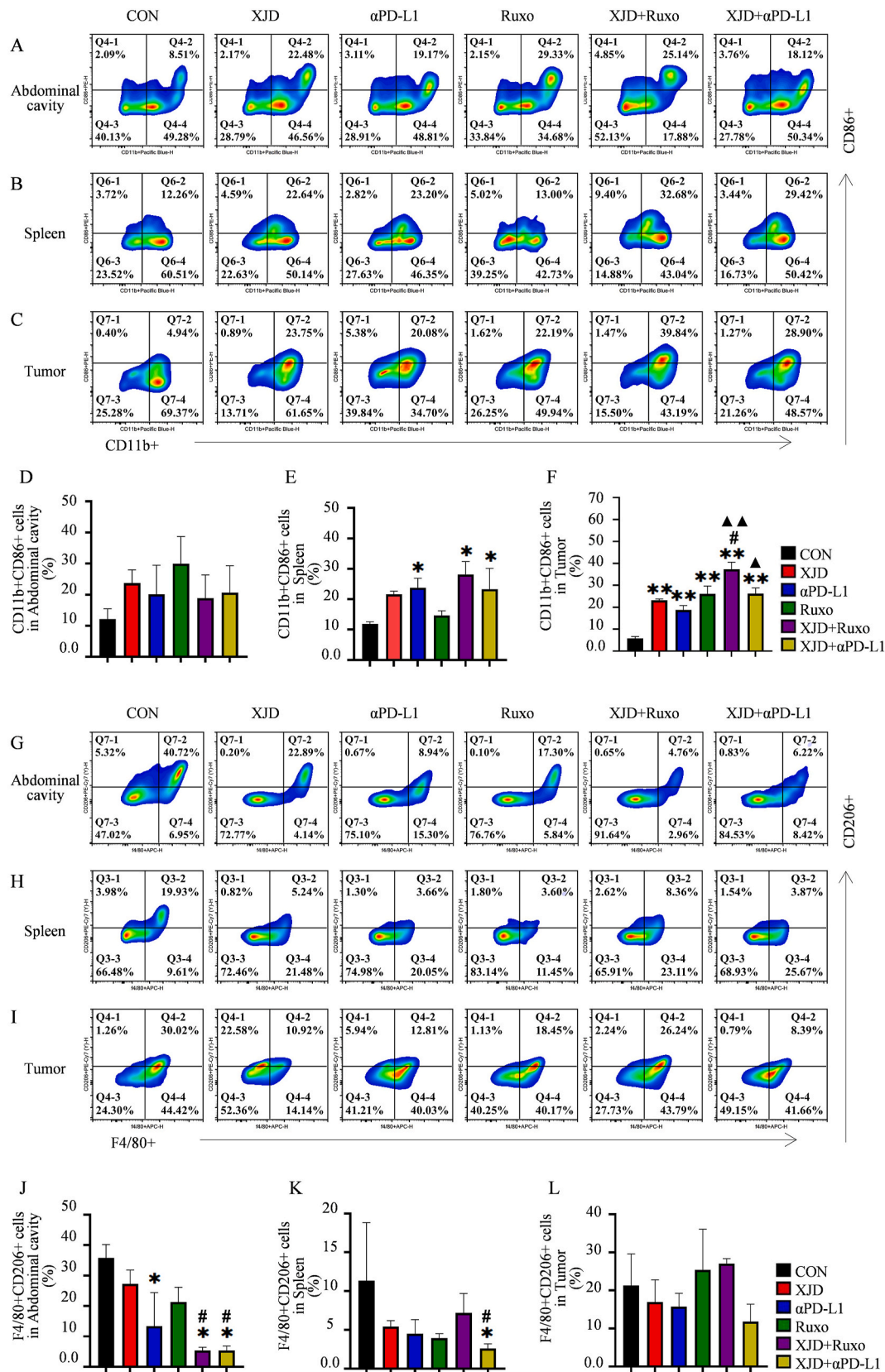


Fig. 8. XJD promotes M1 macrophages polarized and inhibits M2 macrophages polarized *in vivo*. Primary cells were extracted from the abdominal cavity, freshly excised tumors, and spleen for immunological analysis. The cell suspensions were incubated with fluorescently conjugated antibodies specific for CD45, CD206, CD86, CD11b, and F4/80 in the dark for 30 min, then washed with PBS and analyzed using the NovoCyte Quanteon flow cytometer. Flow cytometry data were processed using Novo Express software. Macrophages in the mice, including those from the abdominal cavity, spleen, and tumors, were measured by flow cytometry. **A-F.** M1 macrophages (CD45⁺CD86⁺CD11b⁺) of the LLC model. **G-L.** M2 macrophages (CD45⁺F4/80⁺CD206⁺) of the LLC model. $n = 3$, $\bar{x} \pm s$; * $P < 0.05$, ** $P < 0.01$ compared with control group; # $P < 0.05$, ## $P < 0.01$ compared with XJD group; ▲ $P < 0.05$, ▲▲ $P < 0.01$ compared with αPD-L1 group.

Cell cycle is divided into four phases, including G1, S, G2, and M phases, in mammalian cells, which are controlled by the cyclins and cyclin-dependent kinases (CDKs) (S. Lim & P. Kaldis, 2013). In wild-type p53 (WT p53) cell lines, including A549 cells, studies have shown that the p21 gene induces G2/M phase cell cycle arrest which is mediated in a p53-dependent and p53-independent manner (R. M. de Queiroz et al., 2022; N. Namwan et al., 2022; J. T. Pai et al., 2021). When the expression of the p21 gene was up-regulated, the phosphorylation of Cdk1 and Cyclin B was down-regulated, or p21 protein binds to the Cdk1-Cyclin B complex and inhibits kinase activity Cdk1 (F. B. Charrier-Savournin et al., 2004). This further inhibits WW domain-containing adaptor protein with a coiled-coil (WAC) phosphorylation and its interaction with Aurka, resulting in the inhibition of Polo-like kinase 1 (Plk1) activation and phosphorylation, and cell cycle arrest in G2/M phase (F. Qi et al., 2018). Our study support that XJD induces G2/M phase arrest in A549 cells, which may occur through both p53-dependent and p53-independent mechanisms. However, the p53-dependent effect is absent in p53 mutant cells, such as H1975 cells (M. Nenkov et al., 2020). Our study supports that XJD induced a G0/G1 phase arrest in H1975 cells which is associated with independently of p53.

Inducing apoptosis in tumor cells is one of the approaches used in cancer therapy. (P. N. Kelly & A. Strasser, 2011; Nössing & Ryan, 2023). Apoptosis was first identified as programmed cell death of cells, mediated by cysteine aspartase (N. Van Opdenbosch & M. Lamkanfi, 2019). The Bcl-2 family exerts a regulatory influence on the activity of cysteine aspartase (P. N. Kelly & A. Strasser, 2011). During the process of cell apoptosis, the expression or activity of pro-survival proteins, such as Bcl-2, decreases, whereas the expression or activity of pro-apoptotic proteins, such as BAX, increases (P. N. Kelly & A. Strasser, 2011). This results in the activation of cysteine aspartase, which in turn triggers essential apoptosis-related downstream processes (N. Van Opdenbosch & M. Lamkanfi, 2019). Apoptosis is induced through two distinct mechanisms. One mechanism involves the disruption of the outer mitochondrial membrane, which can result from DNA damage, the accumulation of reactive oxygen species, and other factors (Nössing & Ryan, 2023). This disruption gives rise to the formation of apoptosome (K. Segawa et al., 2014), which in turn leads to the recruitment and auto-cleavage of Caspase-9, as well as the further recruitment and cleavage of Caspase-3 (Julien & Wells, 2017). This cascade of events ultimately promotes apoptosis. To further clarify the anti-tumor mechanism of XJD, we analyzed the apoptosis rate and the expression of apoptotic proteins in the cells. XJD significantly enhances apoptosis in lung cancer cells. These results suggest that XJD may exert its pro-apoptotic effect by decreasing Bcl-2 protein levels, increasing BAX protein expression, and consequently elevating the levels of key apoptotic proteins, specifically Cleaved Caspase-9 and Cleaved Caspase-3. These findings indicated that XJD could serve as an effective therapeutic option for treating neoplastic diseases.

It is evidenced by existing researches that aberrant STAT3 activation is a contributory factor in the formation of tumors (Y. Jiang et al., 2023; H. J. Lee et al., 2014; H. Li et al., 2018a; S. Ouyang et al., 2022). The JAK1/STAT3 pathway has been demonstrated to enhance PD-L1 expression in tumors (C. Liu et al., 2022; M. Shen et al., 2019), inhibit CD8⁺ T cell infiltration (Q. Chen et al., 2022), induce the recruitment of myeloid derived suppressor cells and polarised macrophages (C. Liu et al., 2022), thereby enabling evasion of host immunity, and facilitate the formation of premetastatic niches in the lungs (B. Jing et al., 2020). Blockade of STAT3 signalling in lung tumor cells has been shown to inhibit lung metastasis in immunocompetent mice (B. Jing et al., 2020). Consequently, targeting STAT3 represents a potential strategy for the treatment of NSCLC (K. Shien et al., 2017). XJD has been demonstrated to modulate multiple pathways to inhibit the growth of lung cancer cells. However, it remains unclear whether XJD induces apoptosis in lung cancer cells by affecting the JAK1/STAT3/PD-L1 pathway. The objective of this study was to evaluate the effect of XJD on regulating

JAK1/STAT3/PD-L1 signalling targets, thereby providing key information for the use of XJD in lung cancer treatment. In the present study, we demonstrated that XJD markedly suppressed the expression of the JAK1/STAT3/PD-L1 pathway. Furthermore, intervention with Ruxolitinib, revealed that STAT3 phosphorylation is a key factor in A549 and H1975 cell proliferation. Additionally, Ruxolitinib enhanced the proliferation-inhibitory effects of XJD on lung cancer cells. The combination of Ruxolitinib and XJD further increased apoptosis induction in NSCLC cells and reduced PD-L1 expression in lung cancer cells.

The use of ICIs, particularly those targeting PD-1 and PD-L1, has generated hope among cancer patients worldwide (S. J. Antonia et al., 2018; S. Gettinger et al., 2016). There is a significant correlation between PD-L1 expression in tumor tissues and the effectiveness of PD-1 and PD-L1 inhibitors (J. Zhang et al., 2018). Studies have shown that patients with PD-L1-positive tumors tend to benefit more from these inhibitors (B. Ricciuti et al., 2024). However, recent clinical research indicates that patients with PD-L1-negative tumors may also gain advantages from a combination of ICIs and chemotherapy (S. M. Gadgil et al., 2024). Additionally, evidence suggests that not all patients with PD-L1-positive tumors respond to PD-1/PD-L1 inhibitors (S. M. Gadgil et al., 2024; A. Marcos Rubio et al., 2023). The downregulation of MHC I expression or defects in antigen presentation (J. Tu et al., 2022), decreased sensitivity of tumor cells to interferon- γ (IFN- γ) (J. Tu et al., 2022), and tumor-induced immunosuppression (H. Long et al., 2022) are closely related to resistance to PD-1/PD-L1 inhibitor. It is therefore of great significance to identify strategies of enhancing the efficacy of PD-1/PD-L1 inhibitors.

Animal studies have shown that combining XJD with α PD-L1 significantly inhibits tumor growth in LLC models and increases the spleen index, suggesting an enhancement of immune function in mice. These results indicate that XJD may boost the anti-tumor effects of α PD-L1, supporting the potential clinical application of XJD in conjunction with ICIs. Furthermore, the combination of XJD with Ruxolitinib has been shown to effectively suppress the growth of murine tumors, demonstrating anti-tumor efficacy comparable to that of the combination of XJD and α PD-L1.

The JAK/STAT pathway has been recognized for its critical role in maintaining immune cell homeostasis (N. Fortelny et al., 2024). Furthermore, functional abnormalities or overactivation of the JAK/STAT pathway have been shown to alter the tumor immune microenvironment (Y. Zhang et al., 2022). Previous research has revealed a negative correlation between overactivation of the STAT3 pathway and the infiltration of CD8⁺ T-cells (T. Cao et al., 2024; W. Gao et al., 2021), as well as a positive correlation with immunosuppressive cells such as M2 macrophages (J. Y. Choi et al., 2024; H. Lin et al., 2023). To explore this further, we investigated the macrophages in the mice. Within the tumor immune microenvironment, macrophages can be categorized as either M1 tumor-associated macrophages (M1-TAM), which exhibit a pro-inflammatory phenotype, and M2 tumor-associated macrophages (M2-TAM), which display an anti-inflammatory phenotype (A. Mantovani et al., 2022). M1 macrophages produce pro-inflammatory cytokines such as interleukin 1 beta (IL-1 β), IL-6, and tumor necrosis factor-alpha (TNF- α), supporting pro-inflammatory responses and the modulation of anti-tumor immunity (H. Zhao et al., 2021). In contrast, M2 macrophages secrete anti-inflammatory cytokines like IL-10, leading to anti-inflammatory responses that foster immunosuppression and tumor progression (H. Zhao et al., 2021). Our findings demonstrated that the combination of XJD with α PD-L1 or Ruxolitinib significantly decreased the infiltration of M2 macrophages while notably increasing the infiltration of M1 macrophages in mice. These results imply that XJD may enhance the efficacy of PD-L1 inhibitors by altering macrophage polarization and modulating immunosuppression in mouse models.

In conclusion, the findings of this study demonstrated that XJD inhibited the proliferation of lung cancer cells, induced apoptosis in NSCLC cells, and enhanced the efficacy of anti-PD-L1 therapy by inhibiting the JAK1/STAT3/PD-L1 pathway. These results suggest that

XJD may serve as an effective and safe adjuvant therapeutic agent for lung cancer.

CRedit authorship contribution statement

Chunxia He: Formal analysis. **Yanjing Lu:** Methodology. **Tong Wang:** Software. **Bowen Zhang:** Data curation. **Zuqing Su:** Software. **Lipeng Tang:** Supervision. **Lanting Tao:** Validation. **Guanguan Zheng:** Formal analysis. **Liuning Li:** Funding acquisition. **Bing Feng:** Writing – review & editing. **Ying Zhu:** Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120240>.

Data availability

Data will be made available on request.

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